Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: **BreederNear AI**
b. Team leader name: **Sudarsan N**
c. Problem Statement: **Retail & Commerce: trusted, legal breeder-to-buyer pet commerce in India (BreederNear AI)**

# Brief about the idea

## BreederNear AI: find the trusted breeder next door.

### The problem

- India has about 39 million pets[1], but buying one is a gamble. OLX stopped live-animal listings[2], so the trade moved to WhatsApp and Facebook groups with no checks at all.
- Buyers overpay in shops or lose money to “full advance, courier only” posts with stolen photos.
- Protected native birds are still sold: 300+ of India’s ~1,200 bird species are found in illegal trade[3], and munias are dyed to disguise them[4].
- Real local breeders, often a street away, are invisible and lose hours to manual selling.

### Our solution

- A breeder-to-buyer pet marketplace where every listing is screened by Gemini and by rules in code before any buyer sees it.
- Sellers list from photos and a casual Tamil or English message in under a minute.
- Buyers find trusted pets nearby at fair prices, then get a welfare-sized starter kit and a 14-day care plan.
- “Is this post safe?”: buyers can check any post they saw in a WhatsApp or Instagram group.

**Built from our founder’s five years as a bird breeder and broker in Coimbatore.**

[1] Euromonitor, Pet Care in India (2025) · [2] OLX India: items not allowed · [3] WWF-India / TRAFFIC · [4] Wildlife SOS, Delhi seizure report

# Opportunities

## How is it different?

- Classifieds such as OLX stopped live-animal listings; big pet platforms sell food and accessories.
- A few kennel websites sell dogs, checking only themselves.

**No platform we found checks every listing for wildlife law, registration, scam prices and reused photos before it goes live, or checks posts from outside groups.**

## How does it solve the problem?

- Gemini turns photos and a messy message into a structured, fairly priced listing.
- Trust is decided by transparent code: protected species blocked, dog registration checked, price and scam-language warnings, reused-photo detection.
- Buyers see why: every badge lists its reasons and the questions to ask.
- In-person handover; AI-drafted replies; starter kit and care plan.

## USP

**Trust built in, not bolted on.**

- Protected species blocked from the photo alone: 8/8 in testing, 0/37 false alarms.
- The AI can't override the rules: prompt injection still ends in CAUTION (eval E12).
- Works where the trade happens: Tamil/English messages and outside posts.
- One tool layer for tap screens and the AI chat.

# List of features offered by the solution

## For buyers (customer app)

- Pets near me: trusted first, distance, fair-price bar
- Listing page: trust checks, health notes, questions to ask
- “Which pet suits me?” quiz: home, family, time, budget
- “Is this post safe?” for WhatsApp or Instagram posts
- Contact the breeder; replies appear under Account
- Welfare-sized starter kit, 14-day care plan with vet warning signs
- Local Breeders directory: buy direct at farm prices

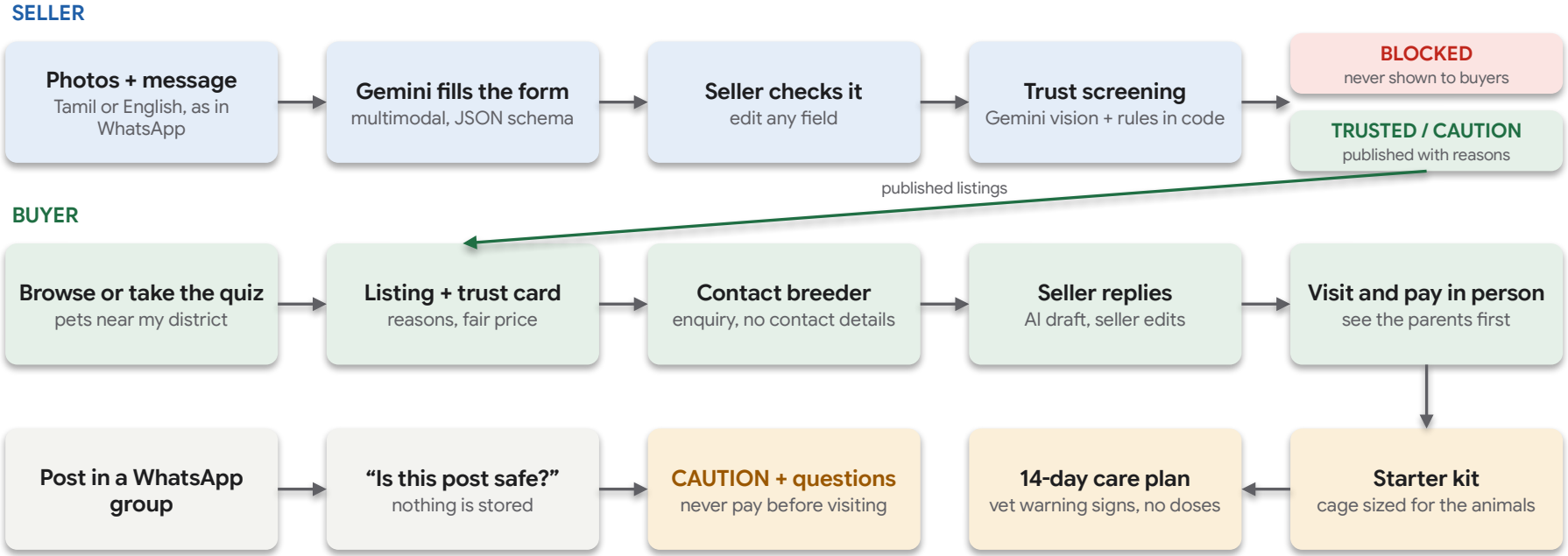
## For sellers (seller app)

- “Fill with AI”: photos + a Tamil/English message become a listing
- Publish with trust check: TRUSTED, CAUTION or BLOCKED, with reasons
- Dashboard: live listings, views, enquiries, trust status
- Manage listings: pause, resume, mark sold
- Enquiries with AI-drafted replies from the listing’s facts
- Farm profile and dog-breeder registration (simulated registry)

## Everywhere

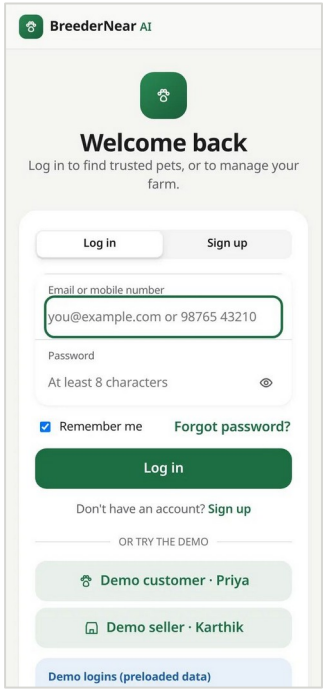
**BreederNear AI chat** (5 Gemini agents, English or Tamil) uses the same tools as the screens. **Safety:** separate customer and seller accounts, roles checked in the API and inside the AI tools, rate limits, AI replies checked for medicine doses. **Accessible:** WCAG 2.2 AA, light and dark, phone to desktop.

# Process flow diagram or Use-case diagram

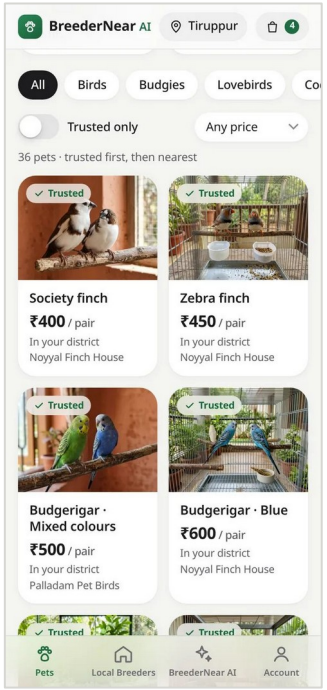



Every step works by tapping, or by asking BreederNear AI in chat: the same tools, the same checks.

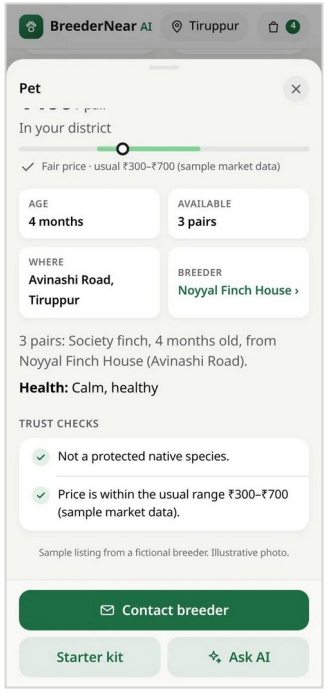
# Wireframes/Mock diagrams of the proposed solution (optional)



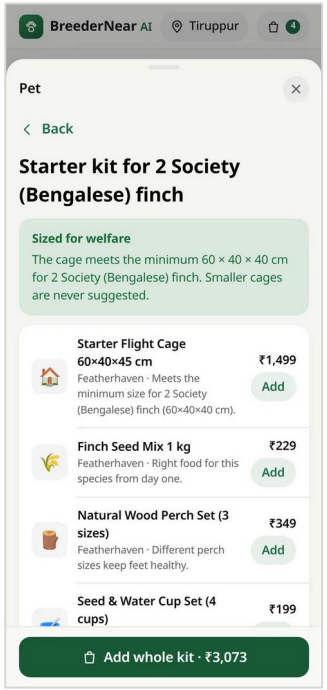

Login: one-tap demo accounts



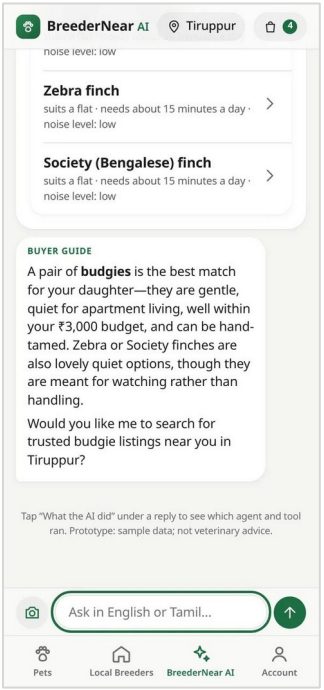

Pets near me




Listing + trust checks

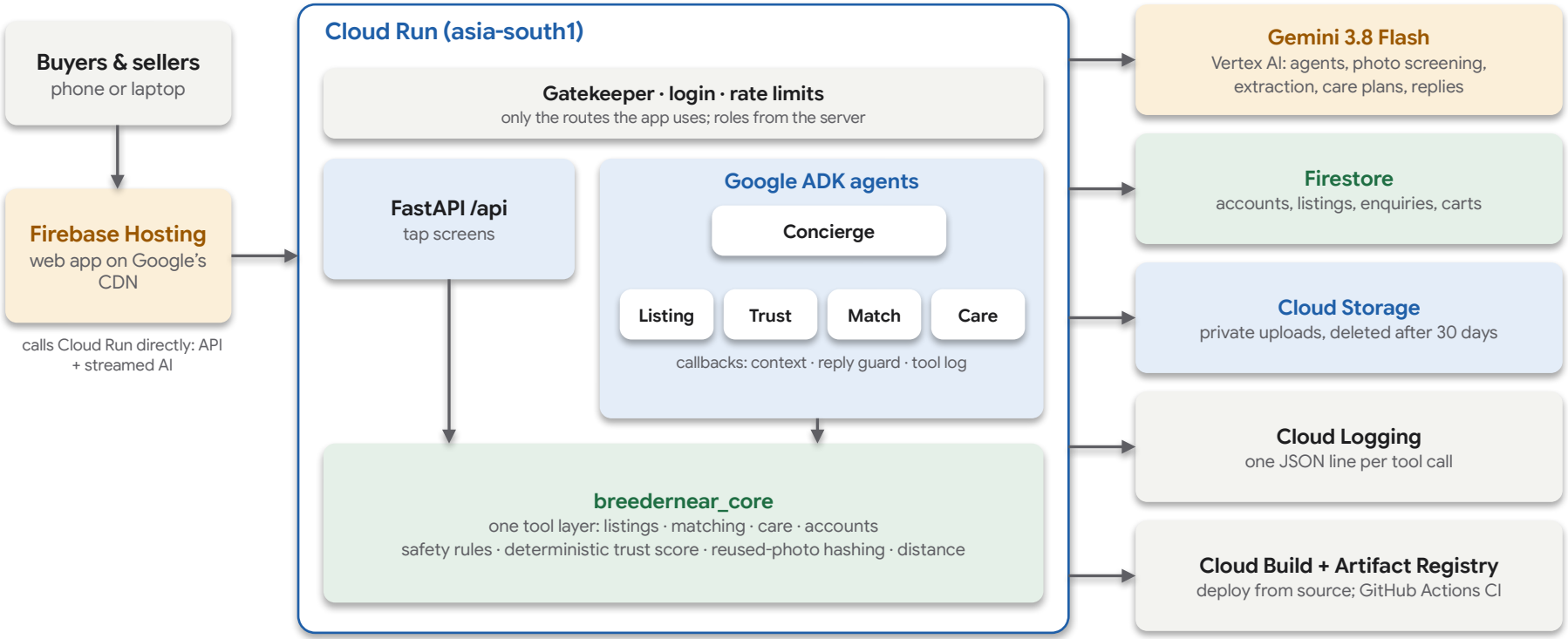



Welfare-sized kit




AI buyer guide

# Architecture diagram of the proposed solution



# Technologies to be used in the solution

| Google technology | How BreederNear AI uses it |
|---|---|
| Gemini 3.8 Flash (Vertex AI) | Five ADK agents; multimodal listing extraction from photos + Tamil/English text; photo screening (species, dye, stock photos); care plans; reply drafts. JSON-schema outputs. |
| Google ADK (Python) | Concierge + listing, trust, match and care agents; tools; callbacks on every agent; sessions; streamed replies |
| Cloud Run | API and agents in one container; login gatekeeper; rate limits; asia-south1 |
| Firebase Hosting | The web app on Google's CDN; calls Cloud Run directly (CORS) so AI replies stream |
| Firestore | Accounts (scrypt-hashed passwords), listings, drafts, enquiries, carts |
| Cloud Storage | Private upload bucket, 30-day deletion |
| Cloud Build, Artifact Registry, Cloud Logging | Deploy from source; structured tool-call logs |
| gemini-3.1-flash-image | Illustrative photos for the fictional sample listings (labelled in the app) |

## Also used

- Python 3.12, FastAPI, Pydantic
- imagehash: perceptual hashing for reused photos
- Plain HTML, CSS and JavaScript: no build step
- pytest, Playwright + axe-core, GitHub Actions
- Firebase CLI (deploy only)

# Estimated implementation cost (optional)

| AI action (measured, real calls) | Gemini tokens | Cost per action* |
|---|---|---|
| List a pet: extract listing + screen photo | ≈ 4,560 in · ≤ 1,075 out | ≈ ₹1.3 |
| Check an outside post (text only; a screenshot adds ≈ ₹0.5) | ≈ 1,620 in · ≤ 445 out | ≈ ₹0.5 |
| 14-day care plan (cached per species and age) | ≈ 560 in · ≤ 1,680 out | ≈ ₹1.2, once per species |
| Reply draft for an enquiry | ≈ 500 in · ≤ 790 out | ≈ ₹0.6 |

| Running cost | Prototype through judging | Example: 10,000 listings a month |
|---|---|---|
| Cloud Run (1 always-on instance) | ≈ ₹600–1,000 / month (estimate) | scales per request |
| Firestore, Cloud Storage, Firebase Hosting | within free usage at prototype scale | pay per use (not measured) |
| Gemini | a few hundred ₹ (tests + judges) | ≈ ₹13,000 / month for screening |

**Screening a listing costs about ₹1.3, against animals priced from ₹400 to ₹25,000 in our sample listings, so trust checks are affordable on every listing.**

*Measured tokens (10 Oct 2026); thinking tokens counted as output · Gemini 3.8 Flash at $1.50 in / $7.50 out per 1M tokens (list price from 1 Jan 2027, as reported by price trackers; introductory price until then is half) · ₹88 = $1 (assumption) · Estimates are labelled

# Business case and go-to-market

## Revenue (planned)

- Verified-breeder subscription: badge, dashboard insights, more listings
- Featured listings in a district
- Commission on starter kits and accessories from partner brands
- B2B: trust-screening API for classifieds and pet shops

## Why it works

- Screening costs about ₹1.3 per listing (measured tokens)
- Serverless: Cloud Run and Firestore cost little when idle and scale per district
- Trust is the product: OLX left this market because it couldn't police sellers

## Go-to-market

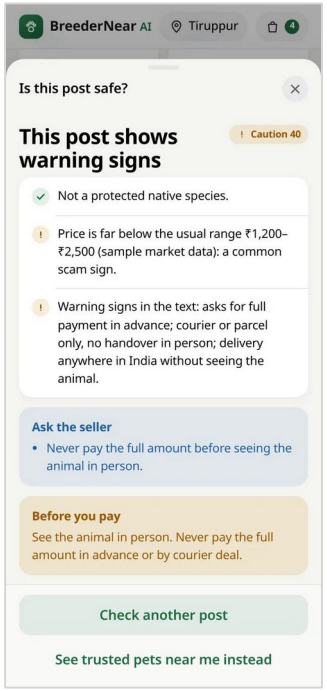
- Start where the founder trades: Coimbatore and Tiruppur breeder groups
- Breeders onboard with one message and their phone photos
- Buyers arrive via "Is this post safe?" links shared in the same groups
- Then district by district across Tamil Nadu

**Who benefits**

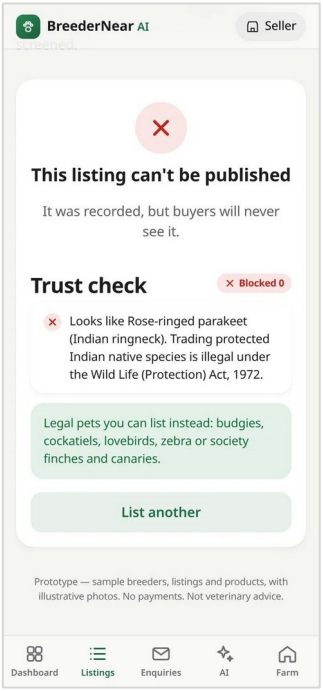
Buyers: fair prices and healthier, legal pets. Breeders: hours saved and buyers beyond their own contacts. Wildlife: fewer protected birds traded, because illegal listings never go live.

Revenue lines and the rollout are plans and assumptions, not results.

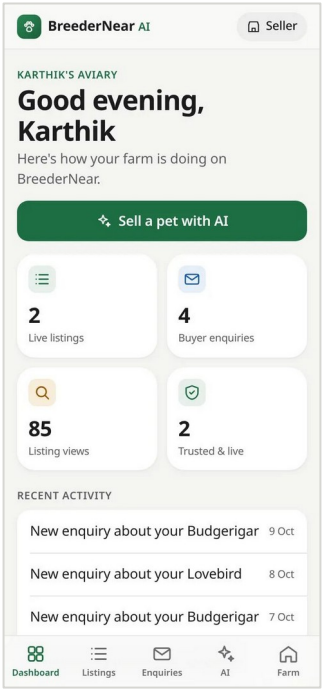
# Snapshots of the prototype



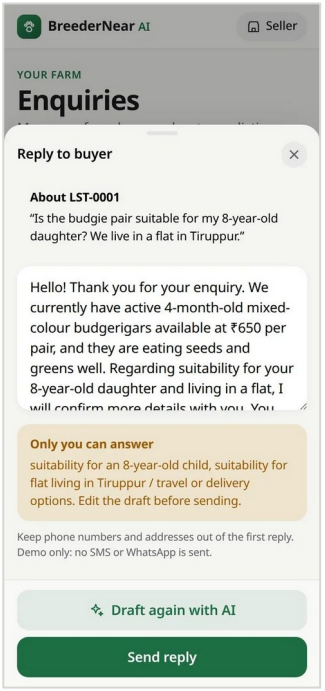

"Is this post safe?"



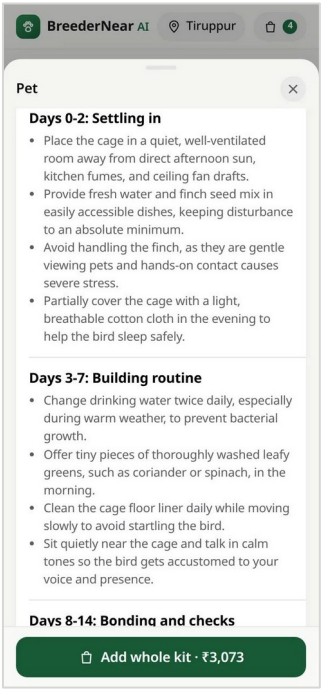

Protected species: blocked




Seller dashboard




AI-drafted reply




14-day care plan

# Prototype Performance report/Benchmarking

| Quality (real Gemini, 10 Oct 2026) | Result |
|---|---|
| Agent evals: 18 cases × 3 runs (listing, Tamil, blocking, scams, reused photos, prompt injection, no medicine advice, roles, explanations, reply drafts) | **54/54 passed** |
| Protected species blocked from the photo alone | **8/8** |
| Ordinary pets wrongly flagged as protected | **0/37** |
| All photo checks: species, dye, blur, no animal, watermark, screenshot | **202/202*** |
| Accessibility: axe-core, WCAG 2.2 AA, light + dark, 320–1280 px | **0 problems** |
| Unit tests in CI · live smoke test | **249 · 17/17** |

| Speed (live) | Time |
|---|---|
| List a pet: upload, AI fill, publish with trust check | 17–27 s |
| Check a pasted post | 5–7 s |
| Web app load (Firebase CDN) | ≈ 0.7 s |
| Gemini: extract a listing | 3–7 s |
| Gemini: screen a photo | 4–5 s |

### How we test

Gemini isn't deterministic, so each eval runs three times and we report the count. Trust decisions are made by tested code, so the model can't be talked out of them.

*Test photos are AI-generated and the screening prompt was tuned on them (first run 196/202), so treat this as an upper bound · Reproduce: scripts/run_evals.sh 3 · scripts/vision_accuracy.py · pytest -m ui tests/ui · evals/RESULTS.md, evals/VISION_RESULTS.md

# Additional Details/Future Development (if any)

## Roadmap (not in the prototype)

- Pilot with real breeders in Coimbatore and Tiruppur, from the founder's network
- List by sending a WhatsApp message (WhatsApp Business Platform)
- Live State Animal Welfare Board and PARIVESH registry checks; the platform registrations India's pet-trade rules require
- Chat history in Agent Engine sessions, so Cloud Run can scale past one instance
- Breeder ratings from buyer feedback and the pet's health after sale
- Hindi, Telugu, Kannada and more districts; vet and pet-pharmacy partners

## Responsible by design (already built)

- Protected species blocked in code, whatever the model says
- No medicine names or doses: checked on every AI reply, even while streaming
- Health signs are "ask the seller", never a diagnosis
- No addresses, ID documents or payment details collected; sellers never see buyers' logins
- Synthetic sample data, clearly labelled; simulated registry numbers start with SIM-

## Provide links to your:

- **GitHub Public Repository**: https://github.com/sudarsan-22/breedernear-ai
- **Demo Video Link (3 Minutes)**: [add the YouTube link on 17 Oct]
- **Final Product Link**: https://breedernear-ai-2026.web.app (API: https://breedernear-655711985039.asia-south1.run.app)

**Try it in 2 minutes: demo accounts with preloaded data**

**Customer:** priya.customer@example.com **Seller:** karthik.seller@example.com **Password:** demo12345

Or tap “Demo customer · Priya” / “Demo seller · Karthik” on the login screen. The README has a step-by-step walkthrough.

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# Thank you

